Reward Function Design for Reinforcement Learning-Based Rocket Landing

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Abstract**—**This paper investigates the effect of reward function design on reinforcement learning performance for autonomous rocket landing. Using Proximal Policy Optimization (PPO) via Stable-Baselines3 in a custom 2D Gymnasium environment, three progressively richer reward formulations are evaluated over 100,000 training timesteps and 20 evaluation episodes per model. Results show that while increasingly complex reward signals produce measurably different trajectory behaviors, none of the three variants achieves a successful landing. Average rewards range from −2095 to −2536, with a 0% success rate across all configurations. Analysis identifies non-directional reward signals, absent terminal bonuses, and insufficient training horizon as the primary failure modes. These findings establish an empirical baseline for minimal-reward PPO performance and motivate structured reward engineering, curriculum learning, and extended training as essential next steps.

Keywords—reinforcement learning; reward shaping; Proximal Policy Optimization; rocket landing; Gymnasium; Stable-Baselines3; reward function design

I. INTRODUCTION

Autonomous rocket landing is a high-stakes control problem of growing practical importance. Aerospace companies such as SpaceX, Blue Origin, and RocketLab now rely on reusable vertical-takeoff, vertical-landing (VTVL) architectures to reduce mission cost and environmental impact. The precision required for pinpoint landing—sub-meter accuracy at near-zero terminal velocity—exceeds the capabilities of classical control methods under uncertainty, making reinforcement learning (RL) an attractive alternative.

RL has demonstrated success in a range of continuous control domains, but its effectiveness is critically dependent on reward function design. A poorly specified reward can produce agents that optimize for the measured signal while failing to achieve the intended task. This problem is especially acute in rocket landing, where the state space is continuous, the dynamics are nonlinear, and the goal state is a narrow target in both position and velocity.

Most prior work addressing RL-based rocket landing relies on multi-stage rewards, imitation learning, or expert-guided curriculum design [1], [2]. These techniques achieve high success rates but obscure the contribution of the reward signal itself by entangling it with policy initialization, curriculum ordering, and expert data. The independent contribution of simple, dense reward formulations—without any of these enhancements—has not been systematically characterized.

This paper addresses that gap. We isolate reward function design as an independent variable and empirically evaluate three progressively richer formulations using PPO in a minimal 2D simulation environment. Our central research question is: *Can a PPO agent learn stable rocket landing using only simple, dense reward signals—without multi-stage rewards, imitation learning, or guidance policies?*

Our results provide a definitive, if cautionary, answer. While each reward variant produces qualitatively distinct trajectory behavior, none converges to successful landing within 100,000 timesteps. This outcome is not a model failure—it is a signal design failure. The findings motivate a set of concrete reward engineering improvements and establish a reproducible baseline against which future work can be benchmarked.

II. RELATED WORK

Su et al. [1] develop a real-time feedback guidance algorithm for 3-DOF pinpoint rocket landing using PPO. Their approach employs a multi-stage reward function that transitions between a guidance-law tracking phase and a free-exploration phase, combined with an imitation learning pre-training framework that initializes policy weights from 600,000 optimal state-action pairs generated by the GPOPS-II convex optimizer. The resulting agent achieves pinpoint landing accuracy (position error < 5 m, velocity error < 2 m/s) with fuel consumption within 6% of the optimal control benchmark, converging after approximately 400,000 training episodes. While highly effective, this methodology entangles reward structure with imitation-based policy initialization, making it impossible to assess the contribution of the reward signal in isolation.

Jiang et al. [2] address the fundamental challenge of sparse reward exploration in a high-fidelity, 6-degree-of-freedom rocket landing model provided by LandSpace

Technology. They introduce the Random Annealing Jump Start (RAJS) algorithm, which uses a baseline PID controller as a guide policy to initialize each episode, then gradually anneals the guide horizon to zero as the RL agent's success rate improves. Starting from an 8% baseline success rate achievable by the PID controller alone, RAJS elevates the PPO agent's success rate to 97% over 10^8 environment steps. Critically, this work demonstrates that standard PPO completely fails (0% success) on the same task without the jump-start mechanism, confirming that sparse or poorly structured reward signals are the primary barrier to convergence in rocket landing control.

Souza et al. [3] implement a 3-DoF longitudinal rocket landing environment using OpenAI Gym and Box2D, and compare PPO against Deep Deterministic Policy Gradient (DDPG) for simultaneous guidance and control. Their shaped reward function combines kinematic state penalties, termination bonuses, and fuel consumption terms. PPO achieves 99.62% successful landings across 5,000 test episodes with 100% success on a 30-trajectory Monte Carlo set, while the DDPG agent achieves only 35.86% success. The study confirms that PPO's on-policy clipped objective and Gaussian exploration provide superior training stability over DDPG in continuous-thrust landing scenarios, and highlights reward shaping as a critical enabler of convergence.

Taken together, these works establish that RL-based rocket landing is achievable, but only with substantial reward engineering, expert initialization, or guided exploration. Each approach relies on at least one of: multi-stage reward curricula [1], imitation pre-training [1], jump-start exploration [2], or shaped termination rewards [3]. The performance of a PPO agent under plain dense rewards—without any of these mechanisms—has not been systematically characterized. Our work fills this gap by evaluating three progressively enriched reward formulations under identical conditions, establishing an empirical lower bound on minimal-reward PPO performance.

III. METHODOLOGY

A. Environment

We implement a custom 2D rocket landing environment using the Gymnasium framework. The simulation models a single-thruster rocket subject to gravitational acceleration of -9.8 m/s² with no drag or wind disturbance. This intentional simplification eliminates confounding variables—attitude dynamics, aerodynamics, and mass variation—to ensure that observed learning outcomes are attributable solely to reward design.

The state vector is four-dimensional: $s = [x, y, v_x, v_y]$, where x and y denote horizontal and vertical position, and v_x, v_y denote the corresponding velocities. The target landing pad is located at the origin (0, 0). Initial conditions are sampled uniformly at the start of each episode: $x_0 \in [-10, 10]$ m and $y_0 \in [50, 100]$ m, with zero initial velocity. The episode terminates on ground contact ($y \leq 0$).

The action space is continuous and one-dimensional: thrust $\in [0, 1]$, applied vertically. No attitude control is included, consistent with our goal of testing minimal control complexity alongside minimal reward complexity.

B. Algorithm and Policy

We train agents using Proximal Policy Optimization (PPO) [3], implemented through the Stable-Baselines3 library. PPO is selected for its stability on on-policy continuous control tasks and its avoidance of replay buffers, which introduce additional hyperparameter sensitivity. The policy network is a multilayer perceptron (MLP) with default Stable-Baselines3 architecture: two hidden layers of 64 units with tanh activation.

All three reward variants are trained with identical hyperparameters to ensure that observed behavioral differences are attributable exclusively to the reward function. Training runs for 100,000 timesteps per variant, and evaluation is conducted over 20 episodes after training concludes.

IV. REWARD DESIGN

We evaluate three reward formulations with increasing signal complexity. Each variant is designed to test the marginal contribution of an additional penalty term, enabling an ablation-style analysis of which reward components drive—or fail to drive—convergence.

Reward V1 provides a pure distance penalty:

$$r_1 = -\text{dist}(s, \text{pad})$$

where dist(s, pad) denotes the Euclidean distance from the current state to the landing pad. This formulation provides a dense signal at every timestep but conveys no information about velocity or trajectory direction.

Reward V2 adds a vertical velocity penalty:

$$r_2 = -\text{dist}(s, \text{pad}) - 0.5|v_Y|$$

The coefficient 0.5 is selected to balance position and velocity penalty magnitudes. This formulation encourages the agent to reduce vertical descent speed as it approaches the ground, with the intent of promoting smoother landing trajectories.

Reward V3 extends V2 with horizontal position and velocity penalties:

$$r_3 = -\text{dist} - 0.5|v_Y| - 0.3|x| - 0.2|v_x|$$

The additional terms penalize lateral displacement and horizontal velocity, encouraging the agent to converge both position axes while maintaining low lateral speed. Coefficients are chosen to maintain approximate balance

among penalty magnitudes given the expected ranges of each state variable.

The design philosophy across all three variants is additive: each version introduces exactly one new signal component. This structure enables direct attribution of behavioral changes to specific penalty terms.

V. EXPERIMENTAL SETUP

Three independent agents are trained, one per reward variant. All training runs use 100,000 timesteps with identical PPO hyperparameters (default Stable-Baselines3 configuration: learning rate 3×10^{-4}, clip range 0.2, n-steps 2048, batch size 64, entropy coefficient 0.0). No reward normalization or observation normalization is applied, to preserve the raw gradient signal produced by each formulation.

Following training, each agent is evaluated over 20 episodes drawn from the same initial condition distribution used during training ($x_0 \in [-10, 10]$, $y_0 \in [50, 100]$). The primary evaluation metrics are: (1) average cumulative episode reward, (2) success rate defined as ground contact within 0.5 m of the pad at $v_y \leq 1$ m/s, and (3) qualitative trajectory behavior as illustrated in Fig. 2.

[Fig. 1: Evaluation reward distributions across 20 episodes for V1, V2, and V3.]

[Fig. 2: Representative agent trajectories for V1 (scattered), V2 (smoother vertical descent), and V3 (more structured, misses pad).]

VI. RESULTS

Table I summarizes the quantitative outcomes across all three reward variants.

Variant	Avg. Reward	Success Rate
V1 (−dist)	−2095	0%
V2 (−dist −0.5|vy|)	−2536	0%
V3 (full signal)	−2535	0%

TABLE I. EVALUATION RESULTS ACROSS REWARD VARIANTS

V1 achieves the highest average reward of −2095, consistent with the fact that it applies the fewest penalties per timestep. V2 and V3 produce average rewards of −2536 and −2535, respectively—lower than V1 despite representing more structurally complete reward signals. This apparent paradox is explained by behavioral analysis: V2 and V3 agents spend significantly more timesteps attempting to modulate velocity before landing, thereby accumulating larger negative reward over longer episode horizons. The reward metric thus reflects episode length as much as policy quality.

No variant achieves a single successful landing across 20 evaluation episodes, yielding a 0% success rate for all configurations. Reward curves for all three variants remain highly variable across evaluation episodes, exhibiting no monotonic improvement trend (Fig. 1). This indicates that none of the agents converged to a stable landing policy within the 100,000-timestep budget.

Qualitative trajectory analysis (Fig. 2) reveals that reward complexity does produce measurably different behavior. V1 trajectories are unstructured—the agent descends with high horizontal variance, frequently landing far from the target. V2 trajectories show smoother vertical descent as the agent learns to reduce $|v_y|$ near the ground, though horizontal dispersion remains high. V3 trajectories are the most structured, exhibiting corrections in both axes, but still fail to converge on the pad location. This progression confirms that each penalty term successfully modifies the relevant state variable's behavior, even in the absence of overall task success.

VII. DISCUSSION

A. Performance Analysis

The trajectory analysis reveals a key tension in reward function design: signal complexity and task success are not the same thing. V3's reward formulation is the most informative—it penalizes four state variables simultaneously—yet it does not produce the best average reward, and it achieves the same zero success rate as V1. The agent responds to each penalty term in a locally sensible way (reducing $|v_y|$, reducing $|v_x|$, correcting lateral drift), but these local behaviors do not compose into a globally convergent landing policy.

This finding is consistent with the broader RL literature on reward hacking and misspecification: an agent can satisfy each individual penalty term while still failing the task, if the reward function does not contain an explicit signal for task completion. In our setting, no version of the reward function assigns a positive signal for reaching the landing pad. The agent therefore has no gradient directing it specifically toward (0, 0)—it has only gradients directing it toward states with smaller penalties.

B. Core Problem: The Reward Signal Gap

The failure of all three reward variants is attributable to three distinct but interacting design deficiencies:

1) Absence of terminal reward: No positive signal is issued upon reaching the landing pad. In the absence of a terminal bonus, the agent has no mechanism to distinguish a successful landing from any other ground-contact episode. The landing pad at (0, 0) is effectively no more attractive than any other ground position, making goal-directed convergence impossible by gradient descent alone.

2) Reward scale versus episode horizon: Cumulative rewards in the range of −2000 to −3500 over episodes of several hundred timesteps produce gradient magnitudes that overwhelm PPO's clipped objective. The per-step signal is small relative to the accumulated negative baseline, reducing the effective signal-to-noise ratio for policy gradient estimation.

3) Complexity without convergence signal: The progression from V1 to V3 adds signal complexity but does not add a convergence direction. V3 trajectories are more structured than V1 trajectories, confirming that the additional penalty terms are processed by the policy network. However, reward does not trend upward across evaluation episodes, indicating that structural behavior change is insufficient for policy convergence in the absence of a directional learning signal.

Collectively, these findings confirm that the observed failure is a reward design failure rather than a model capacity failure. PPO's policy network demonstrably responds to the reward signal—trajectory behavior changes systematically across V1, V2, and V3—but the signal does not contain the information necessary for convergence to the goal state.

VIII. CONCLUSION

This paper presents a controlled empirical investigation of reward function design for RL-based rocket landing. Three dense reward formulations of increasing complexity are evaluated using PPO in a minimal 2D Gymnasium environment. All three variants produce 0% landing success over 100,000 training timesteps, with average episode rewards of −2095, −2536, and −2535 for V1, V2, and V3, respectively.

Despite universal task failure, the results are informative: reward complexity produces measurably distinct trajectory behavior, confirming that the PPO agent processes each penalty term. The failure is attributed to three specific reward design deficiencies—absence of terminal reward, reward scale mismatch, and lack of a convergence gradient—rather than to model capacity or algorithm selection.

Based on this analysis, four concrete improvements are recommended in priority order: (1) addition of a terminal success bonus (+500 for landing within 0.5 m of the pad), which directly supplies the missing goal signal; (2) potential-based reward shaping to provide theoretically-guaranteed policy-invariant gradient enrichment; (3) curriculum learning via staged initial conditions starting close to the pad ($y_0 \in [10, 20]$ m) and expanding progressively; and (4) extended training to 500,000–1,000,000 timesteps, consistent with the scale used in prior successful approaches [1].

This work contributes an empirical baseline for minimal-reward PPO performance on rocket landing, a reproducible environment and protocol for reward ablation studies, and a precise failure analysis that motivates structured reward engineering. The hypothesis that a terminal bonus alone is sufficient to induce convergence under otherwise simple dense rewards remains to be tested and constitutes the immediate next step of this research.

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